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1. The domain of a function is the set of all possible outputs.

1 / 1 point

☒ False

☐ True

✔ Nice work

The domain of a function is the set of all inputs

2. The range of a function f is a subset of its corresponding co-domain ($R_f \subseteq co - D_f$).

1 / 1 point

☐ True

☐ False

✔ Nice work

R_f is the set all images and $co - D_f$ is the set of all possible images, hence, $R_f \subseteq co - D_f$.

3. Which one of the following is the correct domain of the function $f(x) = \frac{1}{x-1}$?

1 / 1 point

☒ $\mathbb{R} - \{1\}$

☐ $\mathbb{R}$

☐ $\mathbb{R}^+$

☐ $[1, \infty)$

✔ Nice work

$f(1)$ is undefined, hence, $D_f = \mathbb{R} - \{1\}$

4. Given a function $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = \frac{x^2-1}{x+1}$, write down the f(1).

1 / 1 point

0

✔ Nice work

$f(1) = \frac{1-1}{2} = \frac{0}{2} = 0$

5. Given a function $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ with $f(x) = \sqrt{x+1}$, which one of the following is the correct set of pre-images of 4?

1 / 1 point

☐ 4

☐ -9, 9

☒ 9

☐ -9

✔ Nice work

$f(9) = \sqrt{9+1} = 3+1 = 4$